Cai, Yifan

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EDUCATION

University of Pennsylvania — **Philadelphia, PA, US**
Ph.D. in Computer & Information Science — *August 2026*
Advisor: Dr. Linh Thi Xuan Phan
Dissertation: Towards Dependable Latency-Sensitive Systems with Timeliness and Security
Research interests: real-time cyber-physical systems, security, and reliability
Master of Science in Computer & Information Science — *August 2024*
Advisor: Dr. Linh Thi Xuan Phan

Shanghai Jiao Tong University — **Shanghai, China**
Bachelor of Engineering, Software Engineering, System Software Track — *June 2020*
Honors/Awards: SJTU Academic Excellence Scholarships (2017, 2018, and 2019)

EMPLOYMENT

University of British Columbia — **Vancouver, BC, Canada**
Postdoctoral Researcher — *September 2026 – Present*
Supervised by Dr. Karthik Pattabiraman and Dr. Aastha Mehta

Roblox — **San Mateo, CA, US**
Research Intern — *May 2024 – August 2024*
Supervised by Dr. Andreas Haeberlen and Dr. Linh Thi Xuan Phan

Alibaba Cloud — **Sunnyvale, CA, US**
Research Intern — *May 2023 – August 2023*
Supervised by Dr. Mo Sha and Dr. Sheng Wang

Alibaba Cloud — **Hangzhou, Zhejiang, China**
Research Intern — *September 2020 – May 2021*
Supervised by Dr. Sheng Wang and Dr. Wenchao Zhou

University of Pennsylvania — **Philadelphia, PA, US**
Research Intern — *July 2019 – December 2019*
Supervised by Dr. Vincent Liu

SELECTED RESEARCH PROJECTS

Security in Industrial Control Systems

- Studying how attackers can infer critical state transitions in industrial control systems (ICS) and launch targeted denial-of-service (DoS) attacks.
- Designing intrusion detection methods to detect these attacks and traffic shaping methods to conceal critical state transitions, while meeting ICS timing requirements.

Security and Privacy in Real-Time Embedded and Cyber-Physical Systems

- Studied how attackers can infer sensitive operational states of real-time embedded and cyber-physical systems through a scheduling side channel.
- Designed an attack that uses only the execution timing and core assignment of an injected observer task to accurately infer states such as whether a user is home and the trajectories of autonomous vehicles.
- Evaluated the attack against a state-of-the-art defense and identified its limitations, including a new vulnerability introduced by the defense.

Security and Reliability in Geo-Distributed Cyber-Physical Systems

- Studied how geo-distributed cyber-physical systems, such as power grids and railway control systems, can remain reliable in the presence of Byzantine faults.
- Designed a protocol that guarantees bounded-time recovery despite unreliable networks and malicious nodes.
- Designed a mechanism to improve availability under attacker-induced packet drops and delays.
- Reduced CPU overhead by a factor of up to 151.8 compared with protocols targeting similar goals.

Memory Security in Latency-Sensitive Systems

- Studied memory security, including confidentiality, integrity, and freshness, in heterogeneous memory systems.
- Designed an efficient encryption scheme and centralized metadata management for encryption and verification, reducing both storage overhead and memory access latency.

PUBLICATIONS AND PREPRINTS

Uncertainty-Aware Resource Allocation for Multi-Path Programs with In-Kernel Predictions
Abigail Eisenklam, Carlos A. Montenegro G., Xian Wang, **Yifan Cai**, Robert Gifford, Linh Thi Xuan Phan, Ricardo G. Sanfelice.
In *38th European Conference on Real-Time Systems (ECRTS)*, 2026

Generative Profiling for Soft Real-Time Systems and its Applications to Resource Allocation
Georgiy A. Bondar, Abigail Eisenklam, **Yifan Cai**, Robert Gifford, Tushar Sial, Linh Thi Xuan Phan, and Abhishek Halder.
In *32nd IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS*), 2026

GeoShield: Byzantine Fault Detection and Recovery for Geo-Distributed Real-Time Cyber-Physical Systems
Yifan Cai and Linh Thi Xuan Phan
arXiv:2511.15031 [cs.CR]

Rasco: Resource Allocation and Scheduling Co-design for DAG Applications on Multicore
Abigail Eisenklam, Robert Gifford, Georgiy A. Bondar, **Yifan Cai**, Tushar Sial, Linh Thi Xuan Phan, and Abhishek Halder.
In *ACM Transactions on Embedded Computing Systems (TECS),*
Special issue of ACM SIGBED International Conference on Embedded Software (EMSOFT), 2025

RoboRebound: Multi-Robot System Defense with Bounded-Time Interaction
Neeraj Gandhi, **Yifan Cai**, Andreas Haeberlen, and Linh Thi Xuan Phan
In *the 20th European Conference on Computer Systems (EuroSys), 2025*

Analysis of Long-term Average Behaviors of Probabilistic Task Systems
Yifan Cai, Linh Thi Xuan Phan, and P.S. Thiagarajan.
In *International Conference on Real-Time Networks and Systems (RTNS), 2024*

Object-oriented Unified Encrypted Memory Management for Heterogeneous Memory Architectures
Mo Sha*, Yifan Cai*, Sheng Wang, Linh Thi Xuan Phan, Feifei Li, and Kian-Lee Tan (* Equal contribution)
In *International Conference on Management of Data (SIGMOD), 2024*

VeriDB: An SGX-based Verifiable Database.
Wenchao Zhou, **Yifan Cai**, Yanqing Peng, Sheng Wang, Ke Ma, and Feifei Li
In *International Conference on Management of Data (SIGMOD), 2021*

Understanding the Effect of Data Center Resource Disaggregation on Production DBMSs.
Qizhen Zhang, **Yifan Cai**, Xinyi Chen, Sebastian Angel, Ang Chen, Vincent Liu, and Boon Thau Loo.
In *46th International Conference on Very Large Data Bases (VLDB), 2020*.

Intra-day Forecast of Ground Horizontal Irradiance Using Long Short-Term Memory Network (LSTM)
Xiuhong Chen, Xianglei Huang, **Yifan Cai**, Haomin Shen, and Jiayue Lu.
In *Journal of the Meteorological Society of Japan*. *Ser. II, 2020, 98(5): 945-957.*

Rethinking Data Management Systems for Disaggregated Data Centers.
Qizhen Zhang, **Yifan Cai**, Sebastian Angel, Ang Chen, Vincent Liu, and Boon Thau Loo.
In *Conference on Innovative Data Systems Research (CIDR)*, *2020*.

Consensus-based Data Statistics in Distributed Network Systems.
Yifan Cai, Jianping He, Wenbin Yu, and Xinping Guan.
In the *57th IEEE Conference on Decision and Control (CDC), 2018*

TEACHING

Teaching Assistant, University of Pennsylvania

- CIS 505 Software Systems, Fall 2021 & Fall 2022
- Delivered lectures on C/C++ programming and debugging with GDB, Valgrind, and related tools to prepare students for coursework.
- Held office hours, graded assignments and exams, and answered questions online.

TALKS

Towards Dependable Latency-Sensitive Systems with Timeliness and Security

- Berkeley Systems Seminar, UC Berkeley, Berkeley, CA, US, February 2026
- Computer Systems Reading Group, University of British Columbia, Vancouver, BC, Canada, September 2026

GeoShield: Byzantine Fault Detection and Recovery for Geo-Distributed Real-Time Cyber-Physical Systems

- Distributed Systems Lab Seminar, University of Pennsylvania, Philadelphia, PA, US, January 2026

Object-oriented Unified Encrypted Memory Management for Heterogeneous Memory Architectures

- Systems Seminar, Yale University, New Haven, CT, US, December 2025
- Distributed Systems Lab Seminar, University of Pennsylvania, Philadelphia, PA, US, October 2025

Analysis of Long-term Average Behaviors of Probabilistic Task Systems

- International Conference on Real-Time Networks and Systems (RTNS), Porto, Portugal, 2024
- Distributed Systems Lab Seminar, University of Pennsylvania, Philadelphia, PA, US, October 2024

Rethinking Data Management Systems for Disaggregated Data Centers

- Distributed Systems Lab Seminar, University of Pennsylvania, Philadelphia, PA, US, December 2019 (with Qizhen Zhang)

Consensus-based Data Statistics in Distributed Network Systems.

- IEEE Conference on Decision and Control (CDC), Miami Beach, FL, US, December 2018

ACADEMIC SERVICE

Conference Reviewer or Program Committee Member

- ACM Conference on Digital Transformation (DXConf): 2026
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS): 2027
- European Conference on Computer Systems (EuroSys): 2027, fall review cycle (reserve reviewer)

Journal Reviewer

- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Internet of Things Journal (IoT-J)
- ACM Transactions on Embedded Computing Systems (TECS)
- Computer Networks, Elsevier

Seminar Organizer

- Distributed Systems Lab (DSL) Seminar: Fall 2025, Spring 2026